

or industrial automation. This tiny chip incorporates everything you need to develop individual projects. Some of its features are:

- The flash memory of high-performance PIC16F676 helps in increasing the processing speed of the microcontroller.
- It comes in PDIP, SOIC, and TSOP packages, all of which are available in 14-pin configuration.
- PIC16F676 contains program memory with memory space around 1.7kB, while RAM and EEPROM memories are of 64 bytes and 128

bytes, respectively.

- One 10-bit ADC module in the device comes with eight analogue channels. This module plays a vital role for sensor interfacing and converting analogue values to digital.
- Power-on reset, comparator, in-circuit serial programming, and master clear reset are some other features incorporated in the device. These help it stay ahead of the other onboard chips and removes the need of buying external components for carrying out different operations. Pin diagram of PIC16F676 is shown in Fig. 4.

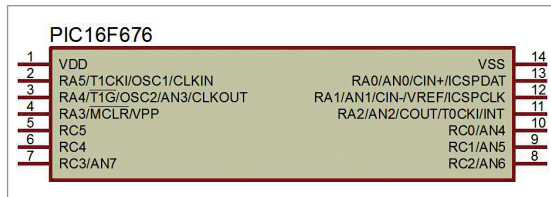


Fig. 4: Pin diagram of PIC16F676

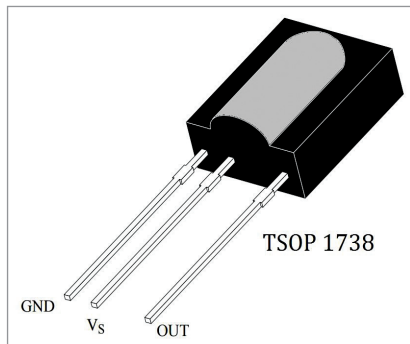


Fig. 5: TSOP 1738 IR receiver

Circuit and working

The project uses PIC16F676 microcontroller and an NEC IR remote to switch on/off any AC load, including lights and fans, from the comfort of your chair or bed. The different IR signals from the remote for the different lights and fans are received by the microcontroller, which then controls the respective relays via a relay driver circuit. These relays are used to switch the lights and fans on or off.

There are many types of IR remote controllers available for different devices, but most of them work around 38kHz frequency. This

TABLE 2
NEC REMOTE LIST
OF CODES

S.No.	Decimal	Hex	Key
1	33441975	1FE448B7	OFF
2	33446055	1FE458A7	MODE
3	33444215	1FE47887	MUTE
4	33446255	1FE4807F	RESUME
5	33449935	1FE440BF	PREVIOUS
6	33442575	1FE4C03F	NEXT
7	33441775	1FE420DF	EQ
8	33444415	1FE4A05F	VOL-
9	33448095	1FE4609F	VOL+
10	33440735	1FE4E01F	0
11	33447695	1FE410EF	RPT
12	33440335	1FE4906F	U/SD
13	33444015	1FE450AF	1
14	33448695	1FE4D827	2
15	33446855	1FE4F807	3
16	33445855	1FE430CF	4
17	33448495	1FE4B04F	5
18	33443615	1FE400FF	6
19	33442175	1FE4708F	7
20	33444815	1FE4F00F	8
21	33442375	1FE49867	9

project requires an ordinary IR TV remote. For detecting the IR signals, TSOP 1738 IR receiver shown in Fig. 5 is used. It can sense the 38kHz frequency signal.

The block diagram of the project is shown in Fig. 6.

In this project the digital IO pins RC0, RC1, and RC2 of Port C of PIC16F676 microcontroller are used to control the relays to switch the appliances on/off. These pins are configured as output pins in the program. The IR receiver is connected to pin RC4 of PIC16F676, which is configured as input pin in the program.

The PIC16F676 microcontroller operates at +5V. Hence, a transformer is used to step down the 230V AC mains supply, which is rectified using a full-wave rectifier. The rectified voltage is regulated to +5V by using regulator IC 7805. The complete circuit diagram is shown in Fig. 7.

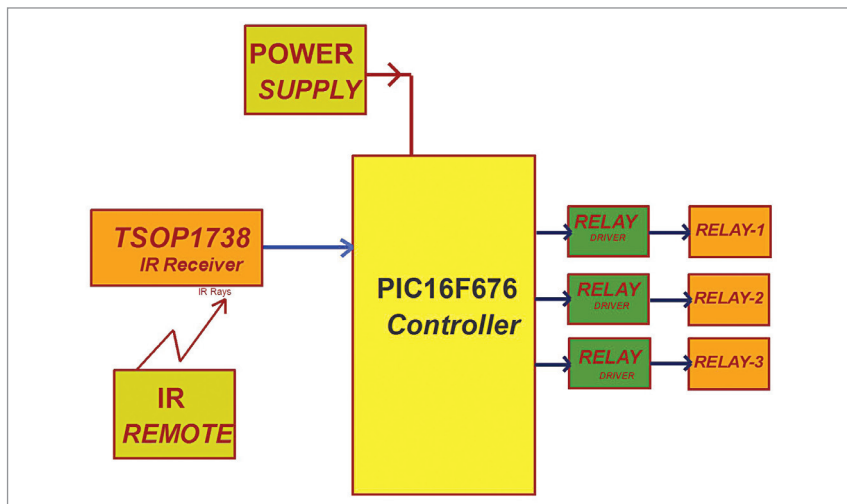


Fig. 6: Block diagram